

3-Screen App with Named Routes & Form Validation

Assignment 7 — Named Routes (Home, Form, Detail) + Registration Form Validation

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Environment: Flutter 3.x / Dart 3.x

Code: https://github.com/SoumitraDeshpande11/assignments/tree/assignments_7

1. Objective

Build a Flutter app with three screens — **Home**, **Form**, and **Detail** — connected with **named routes**. The Form screen is a registration form with real validation: required fields, an email format check, and a minimum-length password rule. On successful validation, the user's data is passed to the Detail screen as route arguments.

2. Requirements Checklist

Requirement	Where it lives
3 screens	HomeScreen, RegisterScreen, DetailScreen
Named routes	MaterialApp(routes: {'/', '/register', '/detail'}) with initialRoute: '/'
Navigation	Navigator.pushNamed(context, '/register') etc.
Data model class	User — immutable class with name and email
Passing data between screens	Navigator.pushNamed(..., arguments: User(...)) read back with ModalRoute.of(context)!.settings.arguments
Required fields	_validateName, _validateEmail, _validatePassword all reject empty input
Email validation	Regex <code>^[\w.+-]+@[\w-]+\.[\w.]+\$</code> checked in _validateEmail
Password validation	Minimum 8 characters, obscured text with a show/hide toggle

3. How Named Routes Work

Instead of building each screen's widget at the call site, the app registers a **route table** in MaterialApp:

```

MaterialApp(
  initialRoute: '/',
  routes: {
    '/': (_) => const HomeScreen(),
    '/register': (_) => const RegisterScreen(),
    '/detail': (_) => const DetailScreen(),
  },
)

```

Any screen can then navigate by name: `Navigator.pushNamed(context, '/register')`. On the web this also maps each screen to a URL (`/#/register`), so screens are deep-linkable.

To send data forward, `pushNamed` accepts arguments. The form passes a `User` object, and the Detail screen pulls it out of the route settings:

```

final user = ModalRoute.of(context)!.settings.arguments as User? ??
_sample;

```

If the screen is opened without arguments (deep link or the "View Sample Detail" button), it falls back to a sample user instead of crashing.

4. How Form Validation Works

The form is a `Form` widget holding a `GlobalKey<FormState>`. Each input is a `TextFormField` with a validator function that returns an error string (shown under the field) or `null` (valid):

- **Name** — required (non-empty after trimming).
- **Email** — required, and must match an email regex (`name@domain.tld`).
- **Password** — required, minimum 8 characters, with an eye-icon toggle to show/hide.

Pressing **Register** calls `_formKey.currentState!.validate()`, which runs every validator. Only if **all** pass does the app navigate to `/detail` with the entered data. The password is never sent to the Detail screen — only name and email.

5. Widget Map

```

MaterialApp (routes table)
├─ '/'          → HomeScreen      ← welcome + 2 navigation buttons
├─ '/register'  → RegisterScreen ← Form[ name, email, password ] + Register
button
└─ '/detail'   → DetailScreen    ← profile card (avatar initial, name,
email)

```

6. Source Code

```
lib/main.dart

1 import 'package:flutter/material.dart';
2
3 void main() => runApp(const RegistrationApp());
4
5 class User {
6   const User({required this.name, required this.email});
7
8   final String name;
9   final String email;
10 }
11
12 class RegistrationApp extends StatelessWidget {
13   const RegistrationApp({super.key});
14
15   @override
16   Widget build(BuildContext context) {
17     return MaterialApp(
18       title: 'Registration App',
19       debugShowCheckedModeBanner: false,
20       theme: ThemeData(
21         useMaterial3: true,
22         scaffoldBackgroundColor: Colors.white,
23         colorScheme: ColorScheme.fromSeed(seedColor: const Color(0xFF2D6A4F)),
24       ),
25       initialRoute: '/',
26       routes: {
27         '/': (_) => const HomeScreen(),
28         '/register': (_) => const RegisterScreen(),
29         '/detail': (_) => const DetailScreen(),
30       },
31     );
32   }
33 }
34
35 // ----- Home
36
37 class HomeScreen extends StatelessWidget {
38   const HomeScreen({super.key});
39
40   @override
41   Widget build(BuildContext context) {
42     return Scaffold(
43       appBar: AppBar(
44         title: const Text('Home'),
45         backgroundColor: const Color(0xFF2D6A4F),
46         foregroundColor: Colors.white,
47       ),
48       body: Center(
49         child: Padding(
50           padding: const EdgeInsets.all(24),
51           child: Column(
52             mainAxisAlignment: MainAxisAlignment.center,
53             children: [
54               const CircleAvatar(
55                 radius: 44,
56                 backgroundColor: Color(0xFFFF4F6F5),
57                 child: Icon(Icons.app_registration,
58                   size: 44, color: Color(0xFF2D6A4F)),
59             ],
60           ),
61         ),
62       ),
63     );
64   }
65 }
```

7. App Screenshots

The app was built for the web (`flutter build web`) and captured in Chrome. Each screen was captured at phone width (500px) via its route URL; the Home screen is also shown at desktop width (1280px).

Home (phone width):



Welcome!

A 3-screen demo using named routes.
Register below to see the Detail screen.

 Open Registration Form

 View Sample Detail

Registration form (phone width):



Register

 Full name

 Email

 Password



 Register

Detail screen (phone width):



Soumitra Deshpande
soumitra@example.com

 Registration complete

 [Back to Home](#)

Home (desktop width):



Welcome!

A 3-screen demo using named routes.
Register below to see the Detail screen.

 [Open Registration Form](#)

 [View Sample Detail](#)

8. Terminal

Real build-and-run session: `flutter pub get`, `flutter analyze` (no issues), `flutter build web`, then `flutter run -d web-server` to serve the app locally.

```
zsh - assignment_7

$ flutter --version
Flutter 3.47.2 • channel stable • https://github.com/flutter/flutter.git
Framework • revision d3b14c8769 (3 weeks ago) • 2026-08-26 16:07:51 -0700
Engine • hash 1cf1c4773fb941c4c74a7f8bb144a8837596c0f4 (revision a804b26164)
Tools • Dart 3.13.2 • DevTools 2.60.0

$ flutter pub get
Got dependencies!

$ flutter analyze
No issues found! (ran in 5.1s)

$ flutter build web --release
Compiling lib/main.dart for the Web... 18.8s
✓ Built build/web

$ flutter run -d web-server --web-port=8743
Launching lib/main.dart on Web Server in debug mode...
Waiting for connection from debug service on Web Server... 10.4s
lib/main.dart is being served at http://localhost:8743

Flutter run key commands.
r Hot reload.
R Hot restart.
h List all available interactive commands.
d Detach (terminate "flutter run" but leave application running).
c Clear the screen
q Quit (terminate the application on the device).
```

9. Run Instructions

`flutter pub get`

`flutter run` # pick a device, or use `flutter run -d chrome`

From Home, tap **Open Registration Form**, fill in name, email, and password, and press **Register**. Try submitting empty fields or a short password to see the inline validation errors. On success, the Detail screen shows the registered user; **Back to Home** pops the whole navigation stack.

10. Possible Extensions

- Add a confirm-password field that must match the password.
- Replace the routes table with `onGenerateRoute` for typed route arguments.
- Show a loading spinner and a success `SnackBar` before navigating.

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